Campus Patrol and Surveillance Drone

Design report: mass budget, power budget and flight time

Beginner problem | Prepared 11 October 2026

Mission and compliance summary

Mission. A small multirotor drone flies a fixed patrol route of about 5 km around the campus perimeter and streams live video to a ground station. It must carry a 250 g gimbal camera payload, cope with wind up to 5 m/s, take off under 3 kg, and have return-to-home and low-battery fail-safes.

Our design. A quadcopter (Quad-X, 4 propellers, about 593 mm wheelbase) on a 6S3P Li-ion 12,600 mAh battery, an ArduPilot Pixhawk 6C Mini flight controller, and a SIYI A8 mini gimbal camera feeding a 5.8 GHz video link. The numbers below are first-pass estimates from momentum theory and manufacturer data and should be confirmed by weighing the parts and flying with logs.

Requirement	Our result
Multirotor	Quadcopter, Quad-X shaped frame
Take-off weight under 3 kg	2,783.4 g, margin 216.6 g (7.2%)
250 g gimbal camera payload	Gimbal camera 95 g; gimbal plus video transmitter 125 g
5 km patrol route	It covers a distance of 5km minimum on both directions of the travel.
Wind up to 5 m/s	Hover tilt 1.0 degrees in a 5 m/s wind. IMU is used to stabilize the tilt in the drone.
Return-to-home and low-battery failsafes	Two-stage battery failsafe, GPS, compass, radio-link failsafes and return-to-home.

Design overview

Patrol path

The route is a fixed set of waypoints around the campus perimeter, loaded as an ArduPilot AUTO mission. The drone climbs to 40 m at 3 m/s, flies the loop at a 10 m/s airspeed (36 km/h) and then lands. The farthest point from home is taken as 2.7 km, which sets the return-to-home energy reserve. The 10 m/s cruise speed sits at the minimum-power point of the power curve (280 W), so endurance is highest there. The tilt is only about 4 degrees, which keeps the camera steady, and the speed is 5 m/s above the wind, so the drone always makes progress into a headwind.

Live video chain

Segment	What it does
On the drone	SIYI A8 mini (3-axis gimbal, 4K camera) and TS5828L 5.8 GHz transmitter (600 mW) and is powered with a 22.V battery
Control and telemetry	ELRS receiver on the flight controller carries pilot commands, and a 433 MHz telemetry radio carries MAVLink to the ground station for mission upload and monitoring.
On the ground	5.8 GHz receiver with a display for the live picture or we can us a laptop if it is available

Component list

Final configuration used for the design; Masses and prices are specified in below table

COMPONENTS	MODEL OF COMPONENTS	PRICE(₹)	WEIGHT(g)
FLIGHT CONTROLLER	Holybro Pixhawk 6C Mini Flight Controller	20459.00	42.4
GPS	Holybro M10 GPS Standard GPS (GHR 1.25mm 10pin)	4500.00	32
BATTERY	Pro-Range INR P42A 22.2V 12600mAh 11C 6S3P Li-ion Drone Battery Pack with XT-90AS	11599.00	1300
MOTOR	T MOTOR NAVIGATOR MN3508 KV380 (X4)	18000.00	412
SPEED CONTROLLER	T Motor V45A LITE 6S 4IN1 ESC 45A	5600.00	13
POWER MODULE	Holybro PM06 V2 14S Analog Power Module & 4-ESC PDB	2031.41	24
BATTERY CONNECTION	XT90 anti-spark lead / adapter for the power module	85.00	15
MANUAL OVERDRIVE	RC receiver (ExpressLRS-type)	1300.00	5
GIMBAL & CAMERA	SIYI A8 mini 4K 3-axis gimbal camera	18996.00	95
LiDAR SENSOR	Benewake TF02-i	8304.00	60
FPV VIDEO TRANSMITTER AND RECIEVER	FPV VTX 1.6W / TS5828L 600mW Transmitter, 5.8G FPV Receiver UVC Video Downlink OTG	606.28	30
WIRING, CONNECTIONS, MOUNTS, FASTENES	MISC	1299.00	75
PROPELLERS	CUSTOM PROPELLERS (X4)	1870.00	60
TELEMETRY	SiK Telemetry Radio V3	6110.00	23.5
CUSTOM FRAME	CARBON FIBER	4000.00	600
TOTAL		**104759.69**	**2786.9**

Power budget

Power by flight phase

Motor power follows the test data (820 g of thrust for 79.9 W, 10.26 g/W) and the design value of 77.0 W per motor in hover, scaled for the real weight. Electronics add 23.85 W.

Phase	Power (W)	Current at 22.2 V	Note
Hover	331.8	14.9 A	307.9 W motors + 23.85 W electronics
Climb at 3 m/s	379	17.1 A	+14% over hover; 13 s to reach 40 m

Phase	Power (W)	Current at 22.2 V	Note
Patrol cruise at 10 m/s	280	12.6 A	15% below hover; minimum-power point
Maximum (all motors at 310 W)	1,264	56.9 A	Worst case, for sizing wires and fuses

Power by supply rail

Rail	Load (W)	Current	Loads
22.2 V battery rail	319.9	14.4 A	Motors (307.9 W) and gimbal camera (12 W)
5.2 V BEC (3 A limit)	11.85	2.28 A (76%)	FC 3 W, GPS 1 W, receiver 1 W, telemetry 1 W, LiDAR 0.85 W, video transmitter 5 W

The video transmitter is the largest load on the 5 V supply, so keep its lead short and add a filter if the video picture shows noise.

Component margins

Check	Rating / limit	Worst case
ESC current per motor	45 A	14.0 A (3.2 times)
Battery current	110 A continuous	56.9 A (1.9 times)
Power module BEC	3 A	2.28 A (76%)
Thrust-to-weight, full battery	2 or more	2.59
Thrust-to-weight at 19.2 V	2 or more	1.93
Power module lead, hover	30 A continuous	14.9 A (2.1 times)
Power module lead, full power	30 A continuous, 60 A burst	56.9 A

The power module lead is rated 30 A continuous. It is fine in normal flight (14.9 A) but not at full power, so use a thicker main lead and a short, thick wire to the ESC.

Battery choice and flight time

Battery choice

Chosen battery: 6S3P Li-ion, 12,600 mAh, 22.2 V nominal (25.2 V full, 19.2 V failsafe), 279.7 Wh, 1.3 kg, 110 A continuous. Six cells in series suit the KV380 motors and 15 in propellers, which turn at about 3,560 rpm in hover and keep the current low (14.9 A). Li-ion cells give a high specific energy (215 Wh/kg), which matters most for endurance. The 3P layout keeps the weight inside the limit and gives 8.7C of discharge rating against a worst-case 4.5C.

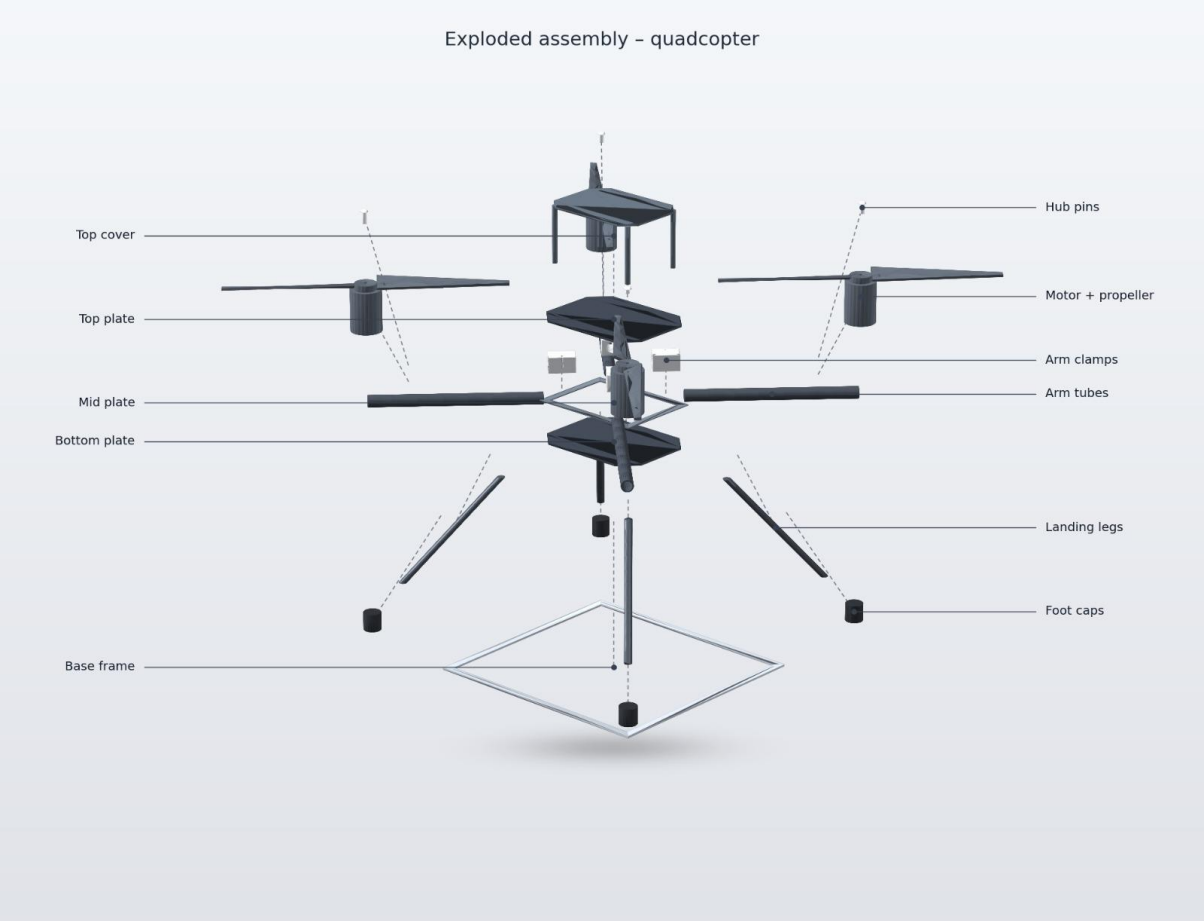
Flight time estimate

Flight time is estimated from battery energy and hover power, then derated for real-world losses.

Step	Calculation	Result
Battery energy	22.2 V x 12.6 Ah	279.7 Wh
Usable energy (80%, keeps 20% reserve)	279.7 x 0.8	223.8 Wh
Ideal hover time	223.8 Wh / 331.8 W x 60	40.5 min

Step	Calculation	Result
Estimated hover flight time (70% derate)	40.5 x 0.7	28.3 min
Estimated patrol flight time (cruise 280 W)	223.8 / 280 x 60 x 0.7	33.5 min
Energy needed for 40 min at cruise	280 W x 40/60 / (0.8 x 0.7)	334 Wh

Camera mounting and antenna placement



Camera

- Mount the gimbal on a vibration-damped plate under the nose, centred on the drone, so the lens sees forward and downward and no propeller or arm enters the picture.
- The gimbal hangs below the frame, so the landing legs must be taller than the lowest point of the gimbal.
- Keep the gimbal power lead and video lead away from the motor phase wires to reduce noise in the picture.

Antennas and sensors

- **GPS and compass:** on a mast at least 10 cm above the frame, at the rear, well away from the power wires and the video transmitter. The IST8310 in this unit is the only compass.
- **5.8 GHz video antenna:** vertical, pointing down below the frame at the rear, with a clear view of the ground station and no carbon plate in the way.

- **ELRS receiver antenna:** vertical, at the front, kept at least 5 to 10 cm away from the video antenna and ESC wires. **433 MHz telemetry antenna:** vertical, on the upper side.
- **LiDAR:** under the centre, facing straight down, with nothing blocking its view.

9. Fail-safes

The flight controller monitors the battery, GPS, compass and radio links. Return-to-home needs a working GPS, so when GPS the drone will try to navigate using magnetometer for direction and land at the next safe point specified by the client. If the drone is unable to go to next safe point it will land at the previous nearest safe point and send SOS signal.

Pre-arm checks block take-off unless GPS lock, compass calibration and battery voltage are good. Return-to-home altitude is 40 m, the same as the patrol altitude, and the last 5 m of landing is flown at 0.5 m/s.

During patrolling, if the battery level is less than required to complete the remaining path, the drone will go to the farthest safe point in path and send critical battery signal

Appendix A. Wiring diagram

Wiring of the components in section 3. Red is the 22.2 V battery rail, orange - the 5.2 V BEC, blue - signal and motor phases, green serial links, purple - analog video and grey - ground. Note: the motor numbering in this diagram differs from the ArduPilot Quad-X order in Figure 1 (M1 front-right CCW, M2 rear-left CCW, M3 front-left CW, M4 rear-right CW); follow Figure 1 when wiring.

